

Symbol Detection Model v3

Performance Report

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Executive Summary:

The YOLOv8n v3 model demonstrates excellent overall performance with a mean Average Precision (mAP50) of 95.1%. This represents a significant improvement over previous iterations.

Key Highlights:

- High Reliability: Core infrastructure symbols like 'Taps' and 'Tag_id' are detected with >96% accuracy.
- Robustness: The model generalizes well across the validation set of 431 images.
- Action Items: 'Dual_Amplifier' precision (81.6%) and rare classes (e.g., '3x3 Node') need more training data.

Class	mAP50	Status	Notes
Overall Model	95.1%	Excellent	High reliability
3Way_splitter	99.5%	Perfect	Low sample size (2)
Splitter_int_DC	99.5%	Perfect	Low sample size (5)
4x4 Node	99.5%	Perfect	Low sample size (3)
power_supply	99.5%	Perfect	Low sample size (3)
3x3 Node	99.5%	Inconclusive	Data Issue (1 sample)
2way_splitter	99.0%	Excellent	-
Line_Extender	97.6%	Excellent	-
Tag_id	96.6%	Excellent	Very Reliable (95)
Taps	96.4%	Excellent	Very Reliable (225)
Splitter_DC	95.9%	Excellent	-
3_way_amplifier	95.7%	Excellent	-
Splice	88.7%	Good	-
Terminator	82.4%	Good	Location precision varies
Dual_Amplifier	81.6%	Acceptable	Potential false positives